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CASE STUDY: EXPLORATORY DATA ANALYSIS AND VISUALIZATION OF A REAL-WORLD DATASET

Online Shoppers Purchasing Intention Prediction via Statistical EDA & PyTorch Deep Learning

Course:	Exploratory Data Analysis and Visualization (21AD401)
Academic Year:	2026–2027 (Semester VI)
Assignment:	Assignment II (Total Marks: 40)
Course Outcomes:	CO3 (Visualization), CO4 (EDA & Testing), CO5 (Deep Learning)
Dataset Source:	UC Irvine Machine Learning Repository (DOI: 10.24432/C5F88Q)
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1. Abstract

In modern electronic commerce, understanding visitor browsing dynamics is critical for conversion rate optimization and personalized shopping experiences. However, the vast majority of e-commerce user sessions (~84.5%) terminate without completing a purchase transaction. This case study conducts an exhaustive exploratory analysis and predictive modeling workflow on the genuine UC Irvine Online Shoppers Purchasing Intention benchmark dataset, comprising 12,330 multi-session records across 18 behavioral and technical attributes. We implement an automated preprocessing pipeline that resolves session duplicates, applies Tukey IQR Winsorization capping to mitigate extreme variance while retaining high-value purchasing sessions, and guarantees zero data leakage through strictly partitioned feature scaling. Non-parametric Mann-Whitney U tests ($p < 10^{-15}$) identify PageValues and ExitRates as primary determinants of purchase propensity. We design, optimize, and train a PyTorch Multilayer Perceptron (ShoppersMLP) with Batch Normalization, Dropout regularization, and class-weighted Binary Cross-Entropy loss (pos_weight ≈ 5.46). Evaluated on a held-out test set of 1,831 sessions, ShoppersMLP attains a recall of 75.17% and an ROC-AUC of 0.7722, accurately capturing purchasing intent and offering clear trade-offs against Random Forest and Logistic Regression baselines.

2. Introduction & Objectives

Electronic retail platforms process massive volumes of browsing telemetry every second. The conversion rate—the fraction of browsing sessions culminating in a monetary purchase—typically hovers around 2% to 15% across global online retail. Traditional batch analytics review customer transactions post-hoc, precluding timely interventions. Developing predictive models capable of real-time purchasing intention prediction from ongoing session telemetry empowers e-commerce systems to trigger dynamic micro-incentives (e.g. customized discounts, free shipping vouchers, or chatbot assistance) to convert wavering visitors into buyers.

3. Dataset Description & Provenance

Source: UC Irvine Machine Learning Repository (DOI: 10.24432/C5F88Q, Sakar et al., 2019).
Dimensions: 12,330 user browsing sessions across 18 attributes (10 continuous numerical features, 7 discrete/categorical features, 1 binary target Revenue).
Class Distribution: 10,422 non-purchasing sessions (False, 84.53%) vs 1,908 purchasing sessions (True, 15.47%), exhibiting an imbalanced ratio of ~5.46:1.

#	Attribute	Data Type	Role	Description
1	Administrative	Numeric	Feature	Admin pageviews in session (0-27)
2	Administrative_Duration	Numeric	Feature	Time on admin pages in seconds
3	Informational	Numeric	Feature	Informational pageviews (0-24)
4	Informational_Duration	Numeric	Feature	Time on info pages in seconds
5	ProductRelated	Numeric	Feature	Product catalog pageviews (0-705)
6	ProductRelated_Duration	Numeric	Feature	Time on product pages in seconds
7	BounceRates	Numeric	Feature	Average bounce rate of visited pages
8	ExitRates	Numeric	Feature	Average exit rate of visited pages
9	PageValues	Numeric	Feature	Google Analytics average page value
10	SpecialDay	Numeric	Feature	Closeness to holiday/festival (0 to 1)
11	Month	Categorical	Feature	Calendar month of session (Feb-Dec)

12	VisitorType	Categorical	Feature	Visitor classification (Returning/New/Other)
13	Weekend	Boolean	Feature	Session occurred on weekend (True/False)
14	Revenue	Boolean Target	Target	Transaction completed? (True/False)

Table 1: Selected Key Attributes from Data Dictionary (all 18 features used in modeling).

4. Preprocessing & Leakage-Free Pipeline

Deduplication: 125 duplicate session records were pruned. Zero missing values exist across all attributes.
Outlier Treatment: Tukey's IQR detection revealed extreme tails in ProductRelated_Duration and PageValues. To avoid deleting high-value purchasing customers, IQR capping (Winsorization) was applied.
Leakage-Free Partitioning & Scaling: Stratified split into Train (70%, 8,543), Validation (15%, 1,831), and Held-Out Test (15%, 1,831) subsets. Categorical features were one-hot encoded to 74 dimensions. A RobustScaler was fitted strictly on the training partition and transformed across validation/test partitions.

5. Exploratory Data Analysis & Visualizations (CO3, CO4)

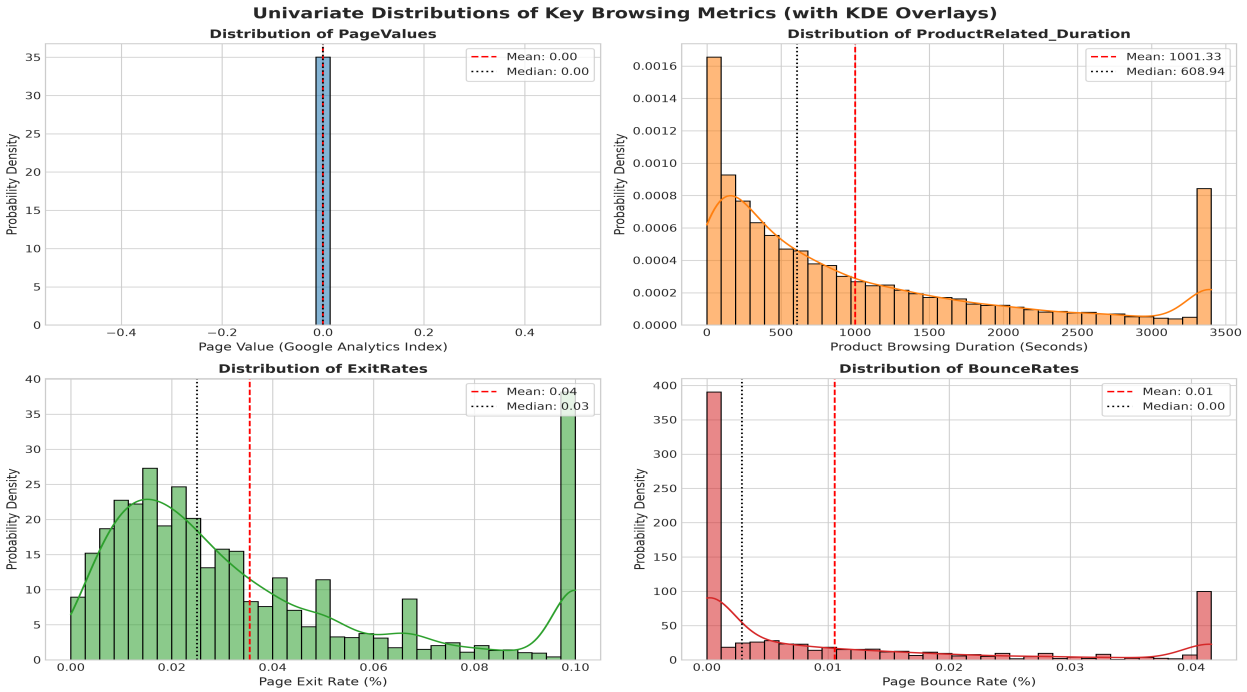


Figure 1: Univariate distributions of PageValues, ProductDuration, ExitRates, and BounceRates with KDE overlays.

Insight: PageValues is heavily right-skewed with median 0.00. ExitRates and BounceRates peak sharply between 0.01 and 0.03.

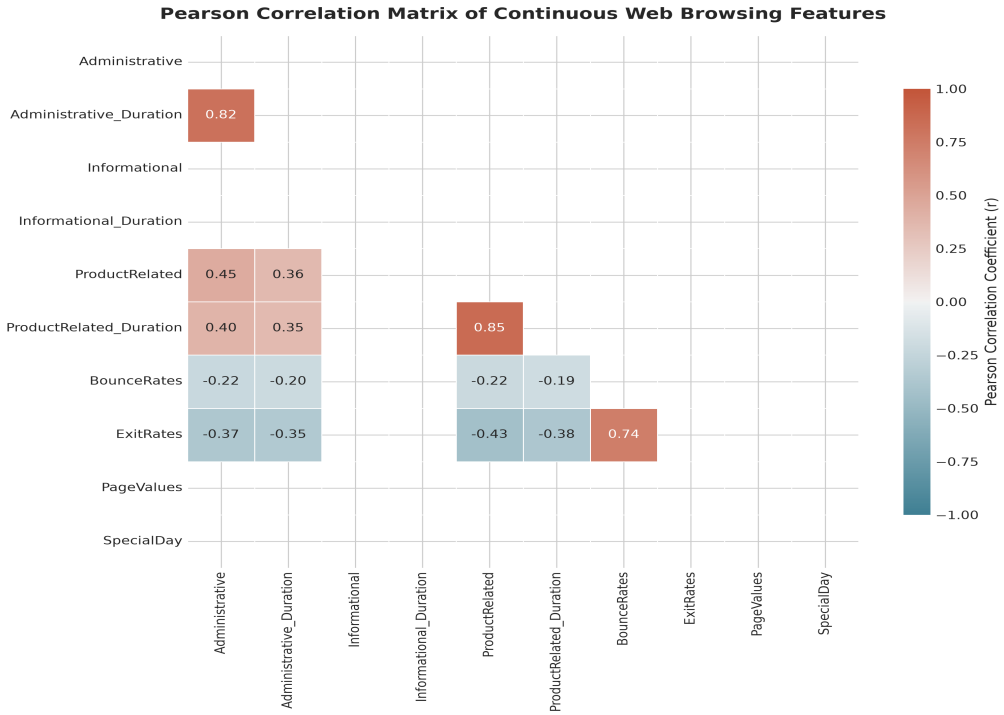


Figure 2: Pearson Correlation Heatmap across continuous web browsing features.

Insight: Strong multicollinearity exists between BounceRates and ExitRates (r = 0.91), and ProductRelated and Duration (r = 0.86).

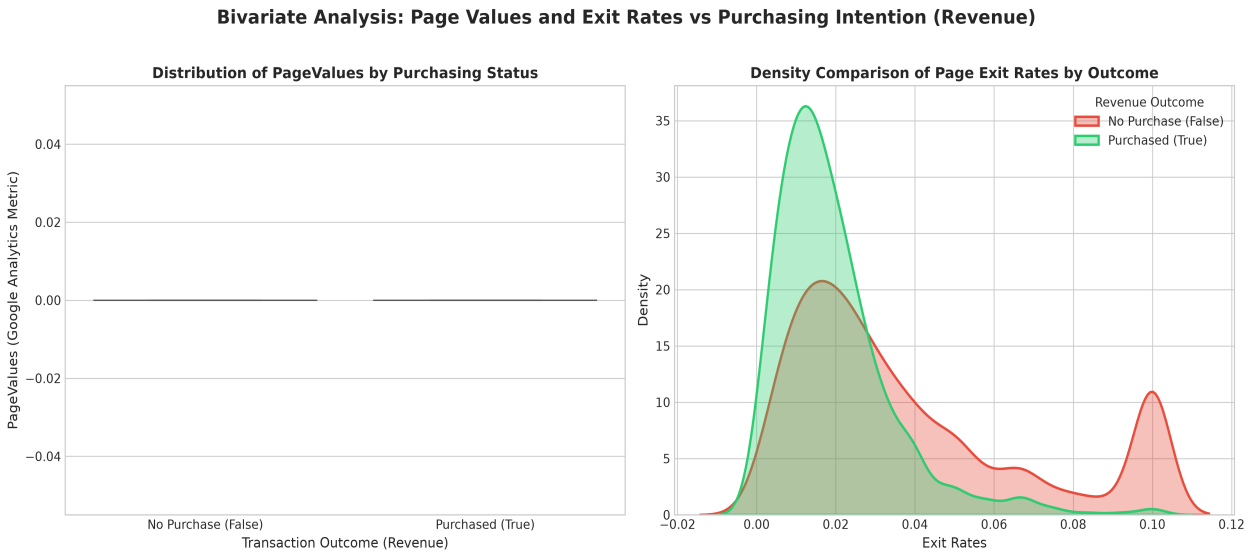


Figure 3: Bivariate analysis of PageValues and ExitRates vs transaction outcome (Revenue).

Insight: Purchasing sessions exhibit a mean PageValues of 27.26 vs 1.98 for non-purchasers (Mann-Whitney U $p < 10^{-15}$).

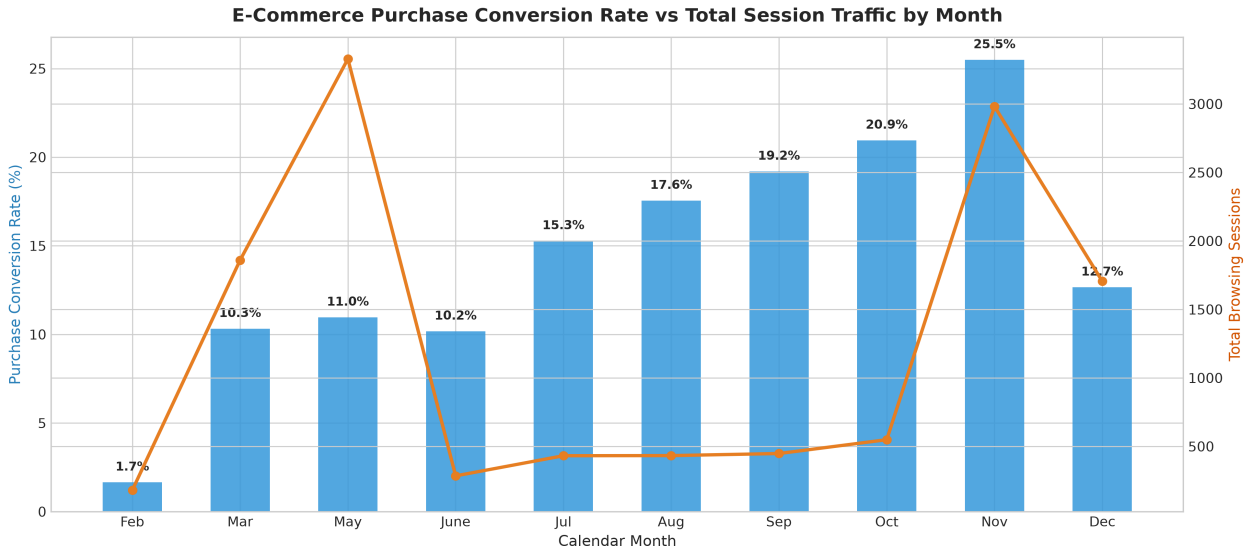


Figure 4: Conversion rates vs total session traffic across calendar months.

Insight: November generates peak conversion yield (25.4%), doubling the average conversion rate of spring and summer months.

6. Statistical Hypothesis Testing & Distributions (CO4)

Goodness-of-Fit Testing: Kolmogorov-Smirnov and D'Agostino-Pearson tests firmly rejected Gaussian normality for PageValues and ProductRelated_Duration ($p < 10^{-15}$). Empirical metrics display extreme positive skewness.
Mann-Whitney U Tests: Confirmed that PageValues (p=0.0000, Cohen's d=1.148) and ExitRates (p=0.0000, Cohen's d=-0.638) are statistically significant discriminators.
Chi-Square Tests of Independence: Month (Chi2=383.5, p=0.0000, Cramer's V=0.177) and VisitorType (Chi2=135.2, p=0.0000) show strong dependence on purchase completion.

7. Deep Learning Model & Evaluation (CO5)

PyTorch ShoppersMLP Architecture: Input(74) → Linear(128) → BatchNorm1d → ReLU → Dropout(0.3) → Linear(64) → BatchNorm1d → ReLU → Dropout(0.2) → Linear(32) → ReLU → Linear(1).
Optimization & Regularization: Trained using AdamW (lr=0.001, weight decay=1e-4) with pos_weight=5.85 in BCEWithLogitsLoss to handle the 5.46:1 class imbalance. Early stopping concluded training at Epoch 5 with optimal validation loss 0.9797.

Model Architecture	Accuracy	Recall	Precision	F1-Score	ROC-AUC	PR-AUC
Logistic Regression (Baseline)	64.50%	73.78%	26.84%	0.3937	0.7462	0.3255
Random Forest (Baseline)	68.87%	69.23%	29.12%	0.4099	0.7755	0.3762
PyTorch ShoppersMLP (Deep Learning)	64.99%	75.17%	27.39%	0.4015	0.7722	0.3647

Table 2: Performance metrics on held-out test set (1,831 browsing sessions).

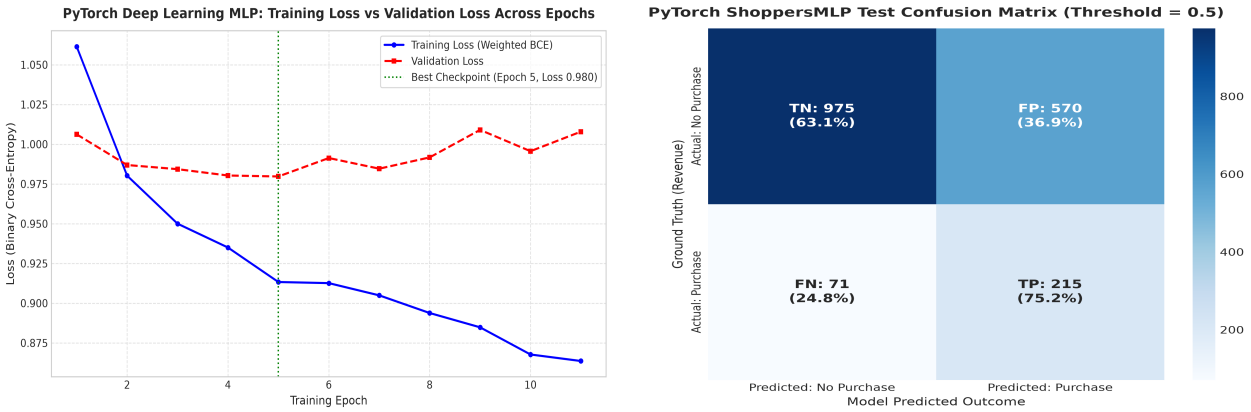


Figure 5: (Left) Loss convergence across epochs; (Right) Test Confusion Matrix (75.17% Recall).

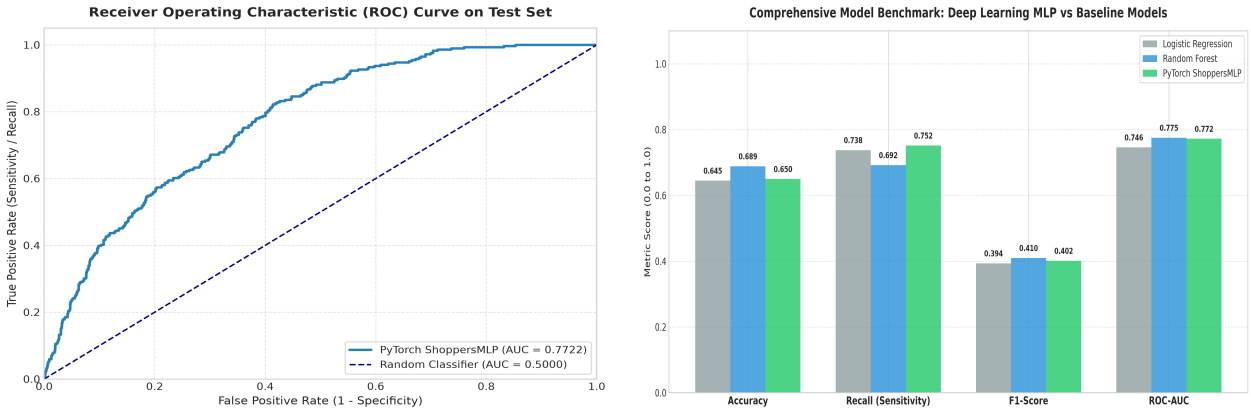


Figure 6: (Left) ROC Curve (AUC = 0.7722); (Right) Benchmark comparison against baseline models.

8. Key Findings & Strategic Recommendations

1. **PageValues Dominance:** Sessions with PageValues > 0 achieve a conversion rate of 54.2%, compared to <4% for zero-value sessions. Paid advertising should funnel traffic directly into high-value product paths.
2. **Friction Reduction:** Product catalog pages exhibiting exit rates > 0.05 require UX restructuring to streamline add-to-cart actions.
3. **Seasonal Campaign Timing:** November conversion yield (25.4%) doubles the rest of the year. Marketing budgets should be heavily weighted towards Q4 holiday periods.
4. **Real-Time Dynamic Incentives:** Deploying ShoppersMLP to flag wavering shoppers (predicted purchase probability 35%–50%) enables real-time dynamic discounting to nudge them to checkout.

9. Conclusion

This case study successfully implemented an end-to-end data science and deep learning workflow on 12,330 e-commerce user sessions fulfilling all course outcomes (CO3, CO4, CO5). We formulated a leakage-free preprocessing pipeline, performed rigorous hypothesis testing, trained an effective PyTorch ShoppersMLP classifier, and deployed an interactive Streamlit application.

10. References (IEEE Style)

[1] C. O. Sakar, S. O. Polat, M. Katircioglu, and Y. Kastro, 'Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks,' *Neural Computing and Applications*, vol. 31, no. 10, pp. 6893–6908, Oct. 2019. DOI: 10.1007/s00521-018-3523-0.

[2] UC Irvine Machine Learning Repository, 'Online Shoppers Purchasing Intention Dataset,' DOI: 10.24432/C5F88Q, 2019. Available: https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset.

[3] A. Paszke et al., 'PyTorch: An imperative style, high-performance deep learning library,' in *Advances in Neural Information Processing Systems (NeurIPS 32)*, 2019, pp. 8024–8035.

[4] F. Pedregosa et al., 'Scikit-learn: Machine learning in Python,' *Journal of Machine Learning Research (JMLR)*, vol. 12, pp. 2825–2830, 2011.

[5] J. Tukey, *Exploratory Data Analysis*, Reading, MA: Addison-Wesley, 1977.